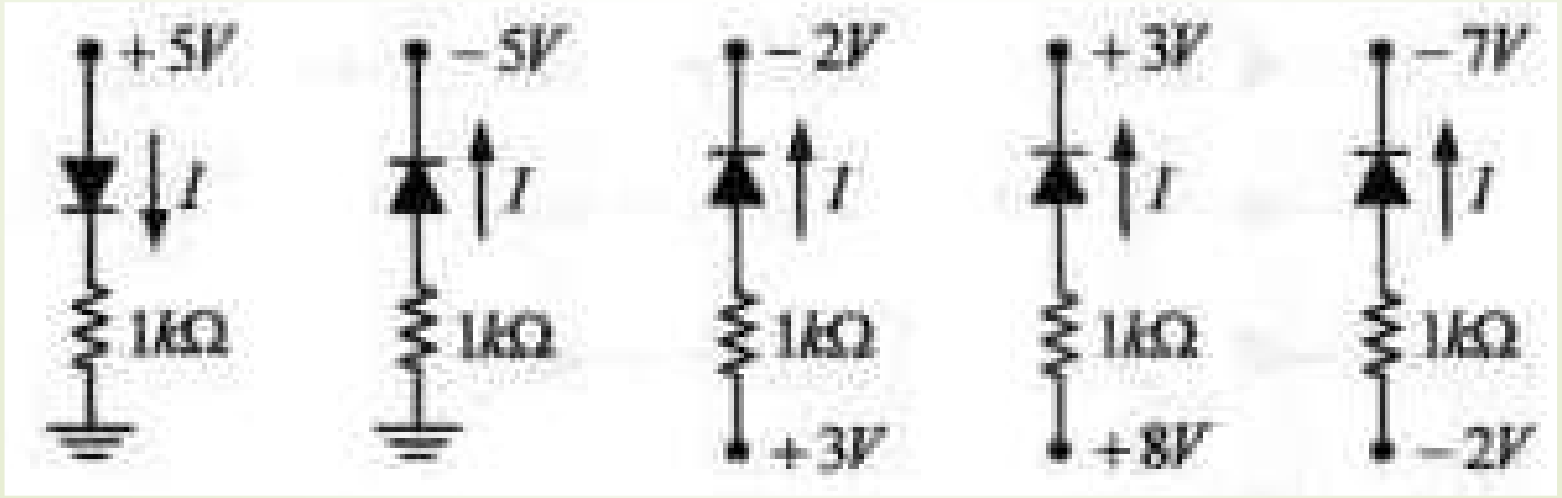
Diode Switching Circuits

Basic Concepts:

Diode switching circuits typically contain two or more diodes, each of which is connected to an independent voltage source. Understanding the operation of a diode switching circuit depends on determining which diodes, if any, are forward biased and which, if any, are reverse biased. The key to this determination is remembering that is **a diode is forward biased only if its anode is positive with respect to its cathode** (see Fig.). One of the very import applications of diode switching circuits is diode logic circuits AND/OR gates.



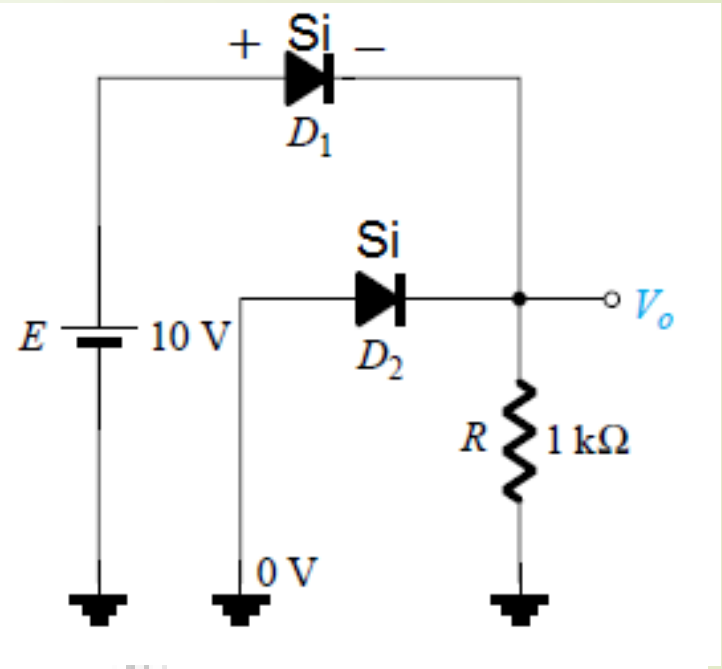
Logic Gates (AND/OR Gates):

Diodes can be used to form logic gates, which perform some of the logic operations required in digital computers.

OR gate: is such that the output voltage level will be a **1** if either or both input is a **1**. The 10V level is assigned a **1** for Boolean algebra while the 0V input is assigned a **0**

Input voltage		State of diodes		Output voltage
V_A	V_B	D_1	D_2	V_O
0	0	off	off	0
0	1	off	on	1
1	0	on	off	0
1	1	on	on	1

Example: Determine V_O for the network in figure.



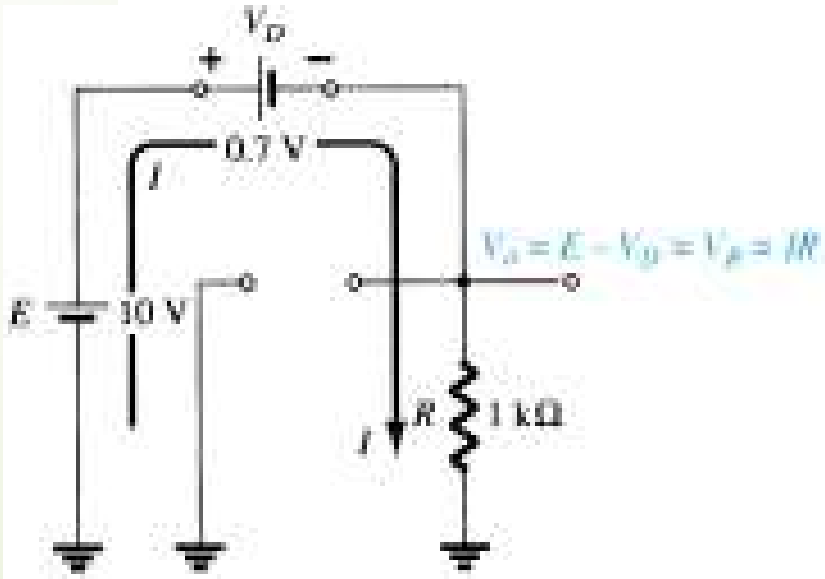
Solution:

D_1 is in the **on** state due to the applied voltage (10V) while D_2 is in the **off** state

$$\mathbf{V_O = E - V_D = 10V - 0.7 = 9.3V}$$

$$\mathbf{= (E - V_D) / R = (10-0.7) / 1k\Omega = 9.3mA}$$

The output voltage level is not 10V as defined for an input of **1**, but the 9.3V is sufficiently at a **1** level with only one input.

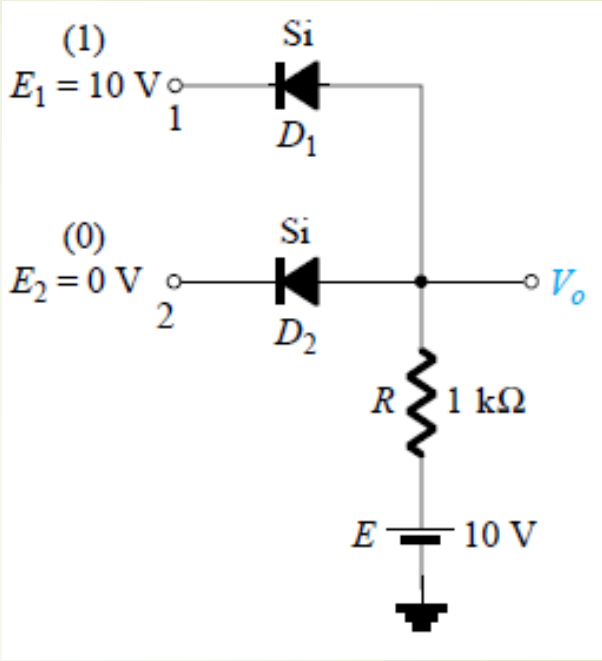


Logic Gates (AND/OR Gates):

AND gate: is such that the output voltage level is will be **1 if both inputs are a 1**.

Input voltage		State of diodes		Output voltage
V_A	V_B	D_1	D_2	V_O
0	0	on	on	0
0	1	on	off	0
1	0	off	on	0
1	1	off	off	1

Example: Determine the output level for the positive logic **AND** gate of figure.

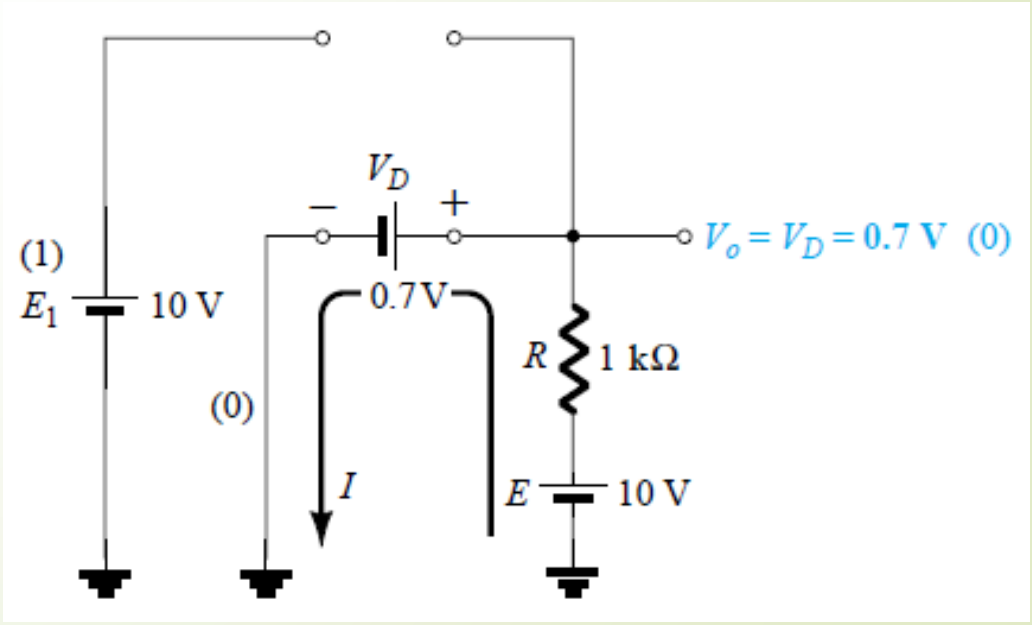


Solution:

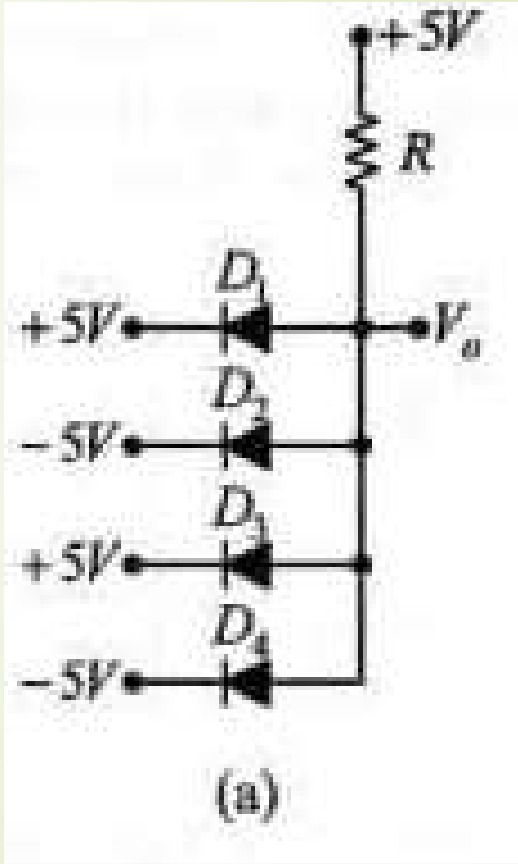
With 10V at the cathode D_1, is assumed that D_1 is in the **off** state.

D_2 is assumed to be in the **on** state due to the low voltage at the cathode side and the availability of the 10V source through 1 KΩ resistor. The voltage at V_O is O.7V due to forward biased diode D2

i.e. $I = (E-V_O) / R = (10 - 0.7) / 1k\Omega = 9.3$ mA



Example: Determine which diodes are forward and which are reverse biased in the circuits shown in figures. Assuming a 0.7 V drop across each forward-biased, determine the output voltage V_O.



Solution:

In (a) the net forward-biasing voltage between supply and input for each diode is

D_1 & D_3: +5 – (+5) = 0 V,

D_2 & D_4: +5 – (-5) = 10 V.

Therefore, D_2 and D_4 are forward biased and D_1 and D_3 are reverse biased.

V_O = -5 + 0.7 = -4.3 V.

Example: Determine which diodes are forward and which are reverse biased in the circuits shown in figures. Assuming a 0.7 V drop across each forward-biased, determine the output voltage V_O.

While in (b) the net forward-biasing voltage between supply and input for each diode is

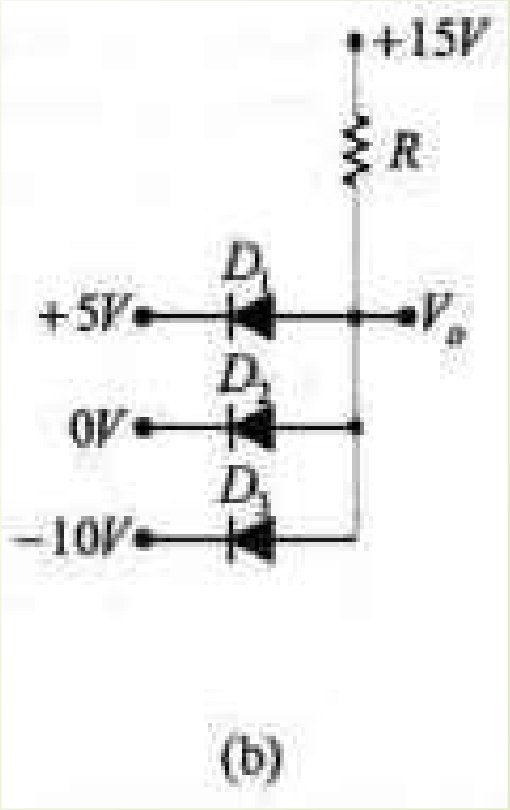
D_1: +15 – (+5) = +10 V,

D_2: +15 – 0 = +15 V,

D_3: +15 – (-10) = +25 V.

Therefore, D_3 is forward biased and D_1 and D_2 are reverse biased.

V_O = -10 + 0.7 = -9.3 V.



Example: Determine which diodes are forward and which are reverse biased in the circuits shown in figures. Assuming a 0.7 V drop across each forward-biased, determine the output voltage V_O.

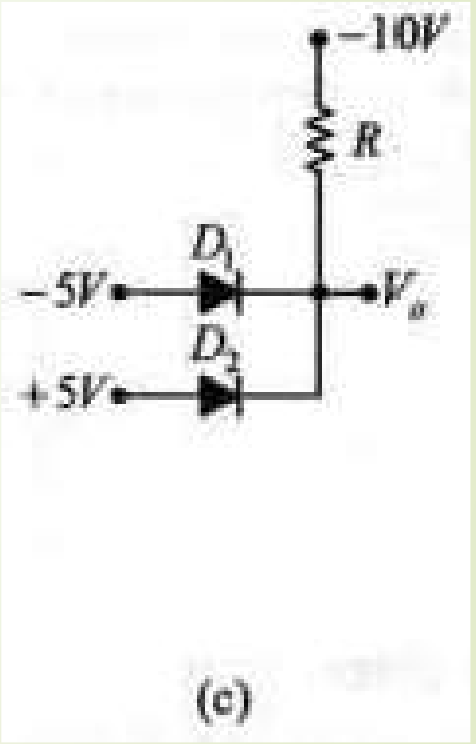
Finally, in (c) the net forward-biasing voltage between supply and input for each diode is

D_1: -5 – (-10) = +5 V,

D_2: +5 – (-10) = +15 V.

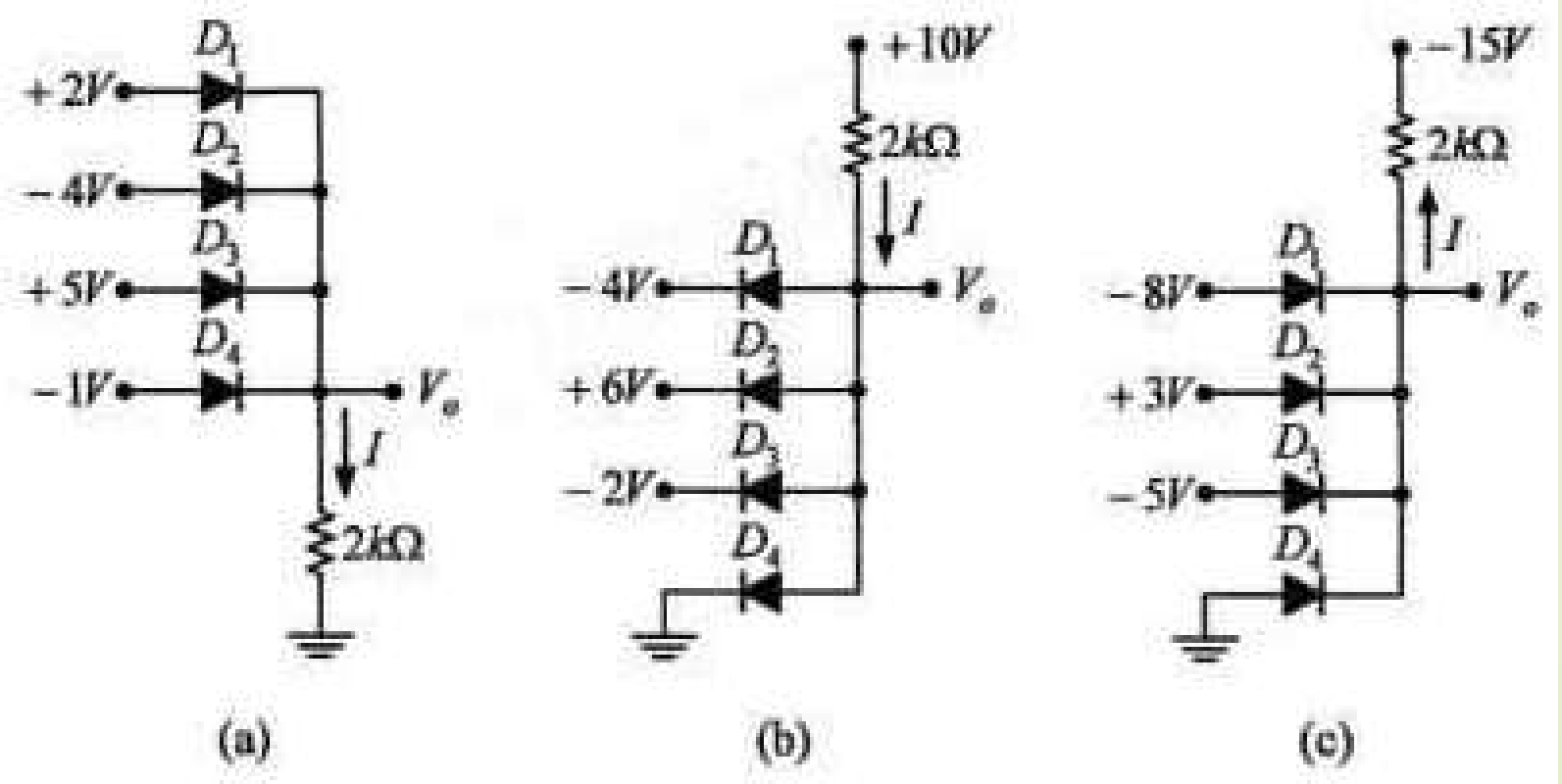
Therefore, D_2 is forward biased and D_1 is reverse biased.

V_O = +5 - 0.7 = +4.3 V.



Exercises:

Determine VO and I for each circuit in figures. Assume that each of the diode in these circuits has a forward voltage drop of 0.7 V.



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Determine VO and I for each circuit in figures. Assume that each of the diode in these circuits has a forward voltage drop of 0.7 V.

